

Question & Answers :

Obtain the inverse Laplace transform of the following :

1. $\frac{4s+6}{s^2+9}$

Solution :

$$\text{Let } F(s) = \frac{4s+6}{s^2+9} = \frac{4s}{s^2+9} + \frac{6}{s^2+9}$$

$$\Rightarrow F(s) = 4L^{-1}\left(\frac{s}{s^2+9}\right) + 6L^{-1}\left(\frac{1}{s^2+9}\right)$$

$$\Rightarrow F(s) = 4L^{-1}\left(\frac{s}{s^2+(3)^2}\right) + 6L^{-1}\left(\frac{1}{s^2+(3)^2}\right)$$

$$\Rightarrow F(s) = 4\cos 3t + 2\sin 3t.$$

2. $\left(\frac{s\sqrt{s}+1}{2s^2}\right)^2$

Solution :

$$\text{Let } F(s) = L^{-1}\left[\left(\frac{s\sqrt{s}+1}{2s^2}\right)^2\right]$$

$$\Rightarrow F(s) = L^{-1}\left[\left(\frac{s^3+1+2s\sqrt{s}}{4s^4}\right)\right]$$

$$\Rightarrow F(s) = \frac{1}{4}L^{-1}\left[\left(\frac{1}{s} + \frac{1}{s^4} + \frac{2}{s^{5/2}}\right)\right], \text{ since } L^{-1}\left(\frac{1}{s^{n+1}}\right) = \frac{t^n}{\Gamma(n+1)}$$

$$\Rightarrow f(t) = L^{-1}[F(s)] = \frac{1}{4}\left[1 + \frac{t^3}{3!} - 2\left(\frac{t^{3/2}}{\Gamma(5/2)}\right)\right] = \frac{1}{4}\left[1 + \frac{t^3}{6} - 8\left(\frac{t^{3/2}}{3\sqrt{\pi}}\right)\right]$$

3. $\frac{s-1}{s^2-2s+5}$

Solution :

$$\text{Let } F(s) = \frac{s-1}{s^2-2s+5} = \frac{s-1}{(s-1)^2+5}$$

$$\Rightarrow f(t) = L^{-1}(F(s)) = 5e^{-t}L^{-1}\left(\frac{s}{s^2+5}\right) - \frac{12}{2}e^{-2t}L^{-1}\left(\frac{2}{s^2+4}\right)$$

$$\Rightarrow f(t) = e^{-2t}(5\cos 2t - 6\sin 2t)$$

$$4. \frac{4s^2-5s+6}{(s+1)(s^2+4)}$$

Solution : Using partial fractions, $\frac{4s^2-5s+6}{(s+1)(s^2+4)} = \frac{A}{s+1} + \frac{Bs+C}{s^2+4}$

$$\Rightarrow 4s^2-5s+6 = A(s^2+4) + (Bs+C)(s+1)$$

Putting $s = -1$, $15 = 5A \Rightarrow A = 3$

Equating coefficient of s^2 on both sides, $4 = A + B$ and hence $B = 4 - A = 4 - 3 \Rightarrow B = 1$

Equating the lowest power of s , $6 = 4A + C$ and hence $C = 6 - 4a \Rightarrow C = -6$

$$\Rightarrow F(s) = \frac{4s^2-5s+6}{(s+1)(s^2+4)} = \frac{3}{s+1} + \frac{s}{s^2+4} - \frac{6}{s^2+4}$$

$$\Rightarrow L^{-1}[F(s)] = L^{-1}\left[\frac{4s^2-5s+6}{(s+1)(s^2+4)}\right] = L^{-1}\left[\frac{3}{s+1} + \frac{s}{s^2+4} - \frac{6}{s^2+4}\right] = 3e^{-t} + \cos 2t - 3\sin 2t$$

5. Find the inverse Laplace Transform using convolution $\frac{1}{(s+2)^2(s-2)}$

Solution :

$$\text{Let } F(s) = \frac{1}{(s+2)^2(s-2)}$$

$$\text{Now, } L^{-1}\left(\frac{1}{(s+2)^2(s-2)}\right) = L^{-1}\left(\frac{1}{(s+2)^2} \cdot \frac{1}{s-2}\right) = te^{-2t} * e^{2t}$$

$$\Rightarrow L^{-1}\left(\frac{1}{(s+2)^2(s-2)}\right) = \int_0^t ue^{-2u} e^{2(t-u)} du$$

$$\Rightarrow L^{-1}\left(\frac{1}{(s+2)^2(s-2)}\right) = e^{2t} \int_0^t ue^{-4u} du$$

$$\Rightarrow L^{-1}\left(\frac{1}{(s+2)^2(s-2)}\right) = e^{2t} \left(u \left(\frac{e^{-4u}}{-4} \right) + \frac{1}{4} \int_0^t e^{-4u} du \right), (\text{integration by parts}).$$

$$\Rightarrow L^{-1}\left(\frac{1}{(s+2)^2(s-2)}\right) = e^{2t} \left(-t \left(\frac{e^{-4t}}{4} \right) - \frac{1}{16} (e^{-4t} - 1) \right)$$

$$\Rightarrow L^{-1}\left(\frac{1}{(s+2)^2(s-2)}\right) = -\frac{te^{-2t}}{4} - \frac{1}{16} (e^{-2t} - e^{2t})$$

$$\Rightarrow L^{-1}\left(\frac{1}{(s+2)^2(s-2)}\right) = \frac{1}{16} e^{2t} - (4t + 1)e^{-2t}$$

6. Solve : $y'' + y = e^{-2t} \sin t, y(0) = 0; y'(0) = 0$.

Solution :

Given $y'' + y = e^{-2t} \sin t$.

Taking Laplace transform on both sides, we get

$$L(y'') + L(y) = L(e^{-2t} \sin t)$$

$$\Rightarrow (s^2 L(y(t)) - sy(0) - y'(0)) + L(y(t)) = \frac{1}{(s+2)^2 + 1}$$

$$\Rightarrow (s^2 + 1)L(y(t)) = \frac{1}{s^2 + 4s + 5} \text{ (Using initial conditions)}$$

$$\Rightarrow L(y(t)) = \frac{1}{(s^2 + 1)(s^2 + 4s + 5)}$$

$$\Rightarrow y(t) = L^{-1} \left[\frac{1}{(s^2 + 1)(s^2 + 4s + 5)} \right]$$

Using partial fractions, $\frac{1}{(s^2 + 1)(s^2 + 4s + 5)} = \frac{As + B}{s^2 + 1} + \frac{Cs + D}{s^2 + 4s + 5}$

$$\Rightarrow 1 = (As + B)(s^2 + 4s + 5) + (Cs + D)(s^2 + 1)$$

On comparing the coefficients of :

$$s^3: A + C = 0$$

$$s^2: 4A + B + D = 0$$

$$s: 5A + 4B + C = 0$$

$$\text{constant: } 5B + D = 1$$

On solving these, we get $A = \frac{-1}{8}; B = \frac{1}{8}; C = \frac{1}{8}; D = \frac{3}{8}$

$$\Rightarrow y(t) = L^{-1} \left(\frac{\frac{-1}{8}s + \frac{1}{8}}{s^2 + 1} + \frac{\frac{1}{8}s + \frac{3}{8}}{s^2 + 4s + 5} \right)$$

$$\Rightarrow y(t) = \frac{1}{8} L^{-1} \left(\frac{s-1}{s^2 + 1} \right) + \frac{1}{8} L^{-1} \left(\frac{s+3}{s^2 + 4s + 5} \right)$$

$$\Rightarrow y(t) = \frac{1}{8} \left(L^{-1} \left(\frac{s}{s^2 + 1} \right) - L^{-1} \left(\frac{1}{s^2 + 1} \right) + L^{-1} \left(\frac{(s + 2) + 1}{(s + 2)^2 + 1} \right) \right)$$

$$\Rightarrow y(t) = \frac{1}{8} \left(\cos t - \sin t + e^{-2t} L^{-1} \left(\frac{s + 1}{s^2 + 1} \right) \right)$$

$$\Rightarrow y(t) = \frac{1}{8} (\cos t - \sin t + e^{-2t} (\cos t + \sin t))$$

7. A beam is simply supported at its end $x = 0$ and is clamped at the other end $x = L$. It carries a load W at $x = \frac{L}{4}$. Find the resulting deflection at any point. The differential equation is $\frac{d^4 y}{dx^4} = \frac{W}{EI} \delta \left(x - \frac{L}{4} \right)$ with boundary conditions $y(0) = y''(0) = 0$ and $y(L) = y'(L) = 0$.

Solution :

The differential equation of deflection is given by $\frac{d^4 y}{dx^4} = \frac{W}{EI} \delta \left(x - \frac{L}{4} \right)$.

Applying Laplace transform on both sides of the equation, we get

$$L(y''''') = \frac{W}{EI} L \left(\delta \left(x - \frac{L}{4} \right) \right)$$

$$\Rightarrow s^4 L[y(t)] - s^3 y(0) - s^2 y'(0) - s y''(0) - y'''(0) = \frac{W}{EI} e^{-\frac{Ls}{4}}$$

Substituting the initial conditions $y(0) = y''(0) = 0$ and assuming

$y'(0) = c_1; y'''(0) = c_2$, we get

$$s^4 L(y(t)) - s^2 c_1 - c_2 = \frac{W}{EI} e^{-\frac{Ls}{4}}$$

$$\Rightarrow s^4 L(y(t)) = \frac{W}{EI} e^{-\frac{Ls}{4}} + s^2 c_1 + c_2$$

$$\Rightarrow L(y(t)) = \frac{W}{EI} \left(\frac{e^{-\frac{Ls}{4}}}{s^4} \right) + \frac{c_1}{s^2} + \frac{c_2}{s^4}$$

$$\Rightarrow y(t) = L^{-1} \left[\frac{W}{EI} \left(\frac{e^{-\frac{Ls}{4}}}{s^4} \right) \right] + c_1 L^{-1} \left(\frac{1}{s^2} \right) + c_2 L^{-1} \left(\frac{1}{s^4} \right)$$

$$\Rightarrow y(t) = \frac{W}{EI} \cdot \frac{\left(x - \frac{L}{4} \right)^3}{3!} u \left(x - \frac{L}{4} \right) + c_1 x + c_2 \frac{x^3}{3!}$$

Therefore, $y(t) = c_1x + c_2 \frac{x^3}{3!}, \text{ for } \left(0 < x < \frac{L}{4}\right)$

$$\Rightarrow y(t) = c_1x + c_2 \frac{x^3}{3!} + \frac{W}{EI} \cdot \frac{\left(x - \frac{L}{4}\right)^3}{6}, \text{ for } \left(\frac{L}{4} < x < L\right).$$
